STS2005(06,07) **미적분학 I(HW1)**

No : _____ 학과 : _____ 학번 : _____ 이름 : _____

1. Use the $\varepsilon - \delta$ definition of a limit to prove:

$\boxed{\text{2.4-\#18}}$ $\lim_{x \to 1}(2x - 5) = -3$

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \in \mathbb{R}$, if $0 < |x - 1| < \delta$, then $|(2x - 5) - (-3)| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|(2x - 5) - (-3)\Big| = \boxed{}\ |x - 1|.$$

Let $\delta = \boxed{}$. Then $\delta > 0$ and for all $x \in \mathbb{R}$, if $0 < |x - 1| < \delta$, then

$$\Big|(2x - 5) - (-3)\Big| = \boxed{}\ |x - 1| < \boxed{}\ \delta = \boxed{} = \varepsilon$$

$\boxed{\text{2.4-\#32}}$ $\lim_{x \to 2} x^3 = 8$

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \in \mathbb{R}$, if $0 < |x - 2| < \delta$, then $|x^3 - 8| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|x^3 - 8\Big| = (x^2 + 2x + 4)\ |x - 2|.$$

If $0 < |x - 2| < 1$, then

$$1 < x < 3 \quad \text{and so} \quad (x^2 + 2x + 4) < \boxed{}, \quad (x^2 + 2x + 4)\ |x - 2| < \boxed{}\ |x - 2|.$$

Let $\delta = \min\{1, \boxed{}\}$. Then $\delta > 0$ and for all $x \in \mathbb{R}$, if $0 < |x - 2| < \delta$, then

$$\begin{aligned}\Big|x^3 - 8\Big| = (x^2 + 2x + 4)\ |x - 2| &< \boxed{}\ |x - 2| \quad (\because |x - 2| < \delta \leqslant 1)\\ &< \boxed{}\ \delta \quad (\because |x - 2| < \delta)\\ &\leqslant \boxed{} = \varepsilon.\end{aligned}$$

$\boxed{\text{2.4-\#37}}$ $\lim_{x \to a} \sqrt{x} = \sqrt{a}$ if $a > 0$.

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \geqslant 0$, if $0 < |x - a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|\sqrt{x} - \sqrt{a}\Big| = \frac{1}{\boxed{}}\ |x - a| \leqslant \frac{1}{\sqrt{a}}|x - a|.$$

Let $\delta = \boxed{}$. Then $\delta > 0$ and for all $x \geqslant 0$, if $0 < |x - a| < \delta$, then

$$\Big|\sqrt{x} - \sqrt{a}\Big| \leqslant \frac{1}{\sqrt{a}}|x - a| < \frac{1}{\sqrt{a}}\delta = \boxed{} = \varepsilon$$